

Aaron Low Weng Soon

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SUMMARY

Senior robotics engineer with six years in autonomous vehicles, specialising in multi-sensor calibration and fusion. I've worked on systems that utilise a combination of deep learning and classical methods like factor graphs, multi-view geometry, and SLAM. I work end to end, developing the algorithms and building the scalable, automated tooling and infrastructure that makes them fast to iterate on and reliable to run. I've built offline systems with abundant compute and onboard systems under tighter compute constraints. I've also led technical direction, mentored junior engineers, and helped shape hiring decisions.

WORK EXPERIENCE

Wayve

2024 - Present

Senior Robotics Engineer (Sensor Calibration and Fusion) • 2024

London, United Kingdom

- Designed and implemented multi-sensor calibration algorithms (camera, IMU, lidar) combining deep learning with multi-view geometry and SLAM.
- Built scalable automated pipelines running calibration continuously supporting a fleet of vehicles spread across multiple locations globally.
 - Our improvements led to a 10x speedup in turnaround time for calibrating a vehicle.
- Built an end-to-end deep learning pipeline for training and evaluating visual feature extraction and calibration models, covering dataset generation, curation, and loading.
- Implemented metrics and automated validation that detect miscalibration and drift, catching issues before they reached downstream users.
- Developed internal tooling to deploy, track, and maintain per-vehicle calibrations across the fleet.

Motional

2020 - 2024

Senior Research Engineer (Sensor Calibration) • 2023 - 2024

Singapore

Research Engineer (Sensor Calibration) • 2020 - 2023

- Developed solutions for sensor (cameras, LIDARs, radars, IMU) calibration and validation of autonomous vehicles.
- Researched deep learning solutions for sensor calibration as well as implementing the infrastructure for training and evaluation.
- Built and maintained cloud-based systems for continuous validation of vehicle sensor data.
 - Our automated system ran daily on a fleet of hundreds of vehicles allowing data-driven calibrations.
- Automated dataset generation and curation for algorithm research.
- Worked on vehicle hardware to deploy online sensor calibration algorithms.

Hyundai-Aptiv Autonomous Driving Joint Venture

2020

Autonomous Vehicle Intern (Sensor Calibration)

Singapore

- Researched deep learning for camera-to-LiDAR calibration and deployed a RegNet based model into production with TensorRT.
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PUBLICATIONS

- UniCal: a Single-Branch Transformer-Based Model for Camera-to-LiDAR Calibration and Validation — a Transformer architecture using self-attention to perform camera-to-LiDAR calibration and validation in a single branch.
- Depth to Colour Translation for 3D Hand Pose Estimation From Monocular RGB With Generative Adversarial Networks — MEng thesis, Imperial College London.

EDUCATION

Imperial College London

2018 - 2019

PhD Researcher, Machine Learning and Computer Vision

- Research area: 3D Pose Estimation

Imperial College London

2014 - 2018

Electrical and Electronic Engineering MEng

- First Class Honours • Dean's List (top 10% of class) Year 2
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TEACHING

HELP University

2019 - 2020

Lecturer, Faculty of Computing and Digital Technology

- Lectured introductory programming and presented deep learning tutorials to both staff and students.
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PROJECTS

Autonomous Snack Delivery Android (ASDA)

2017

Robot Navigation

- Autonomous robot that manoeuvres a building and takes an elevator to obtain and deliver snacks.

ParkWare

2016

Machine Learning and Web

- Prize Winner ("Best use of Amazon Web Services") Parking space detection web service using machine learning to detect cars in parking lots.

Without Abandonware

2021

Unity/C# Developer

- Featured by GitHub in "Top entries from Game Off 2021". A 2D platformer written in Unity with C#.
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TECHNICAL SKILLS

Programming: Python • C++ • C#

ML and Robotics: PyTorch • TensorFlow • ROS • TensorRT • Multi-view Geometry • SLAM • GTSAM

Infrastructure: AWS • Azure • Terraform • Docker • Jenkins • Serverless • Bazel • CMake

Tools and Data: Git • Linux • Bash • Pandas • SQL • Unity

Languages: English (Native) • Malay (Limited Working)
